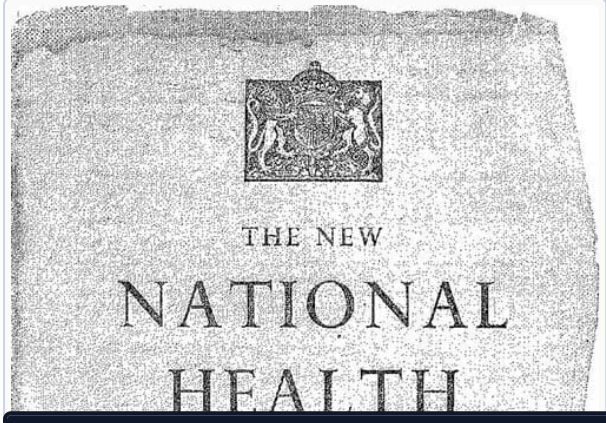
PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 OPTION 11

MODERN MEDICINE IN BRITAIN (C1900–PRESENT)

Genetics, The Antibiotic Revolution & The National Health Service



HISTORICAL OVERVIEW • C1900–PRESENT

The twentieth century witnessed an unprecedented acceleration of clinical capability. In 1928, Alexander Fleming discovered penicillin, transformed into mass-produced antibiotics by Florey and Chain during the Second World War. With the 1948 launch of the National Health Service by Aneurin Bevan, medical treatment in Britain became a universal human right, free at the point of delivery.

From Crick and Watson unravelling the double helix of DNA in 1953 to robotic surgery, organ transplants, and public health drives combating smoking and cancer, medicine advanced from fighting microbes to mastering molecular genetics. This volume traces how state welfare and high-technology science radically extended modern life expectancy.

SPECIFICATION & ENQUIRY DIRECTORY

Pearson Edexcel GCSE (9–1) History

ENQUIRY 01 pp. 3–4

Ideas on Causes: Genetics, DNA & The Human Genome

KEY QUESTION: *"How did unlocking the double helix revolutionize understanding of causes?"*

- **Double Helix Discovery (1953):** Francis Crick, James Watson, Rosalind Franklin, and Maurice Wilkins discovered the molecular structure of DNA.
- **Human Genome Project (1990–2003):** International scientific consortium that successfully mapped all 3 billion chemical base pairs in human DNA.
- **Genetic Diagnosis & Therapy:** Identified hereditary root causes of cystic fibrosis, Down's syndrome, and breast cancer genes (BRCA1 and BRCA2).

ENQUIRY 02 pp. 5–6

Lifestyle Factors & Advanced Diagnostic Technologies

KEY QUESTION: *"How did lifestyle diseases and scanning transform 20th-century medicine?"*

- **Lifestyle Pathology:** Recognition that smoking, alcohol, poor diet, and physical inactivity cause coronary heart disease, diabetes, and cancers.
- **Advanced Medical Imaging:** Diagnostic revolution via mobile X-rays, ultrasound (1950s), CT scans (Godfrey Hounsfield 1972), and MRI scanners (1970s).
- **Point-of-Care Diagnostics:** Blood sugar monitors, endoscopes, and rapid genetic sequencing enabling early micro-level detection before symptoms escalate.

ENQUIRY 03 pp. 7–8

Magic Bullets, High-Tech Treatments & The NHS (1948)

KEY QUESTION: *"How did synthetic magic bullets and the NHS dismantle medical inequality?"*

- **Chemical Magic Bullets:** Paul Ehrlich synthesized Salvarsan 606 (1909) for syphilis; Gerhard Domagk discovered Prontosil (1932) for puerperal sepsis.
- **Foundation of the NHS (1948):** Launched by Health Minister Aneurin Bevan, establishing free healthcare at the point of delivery based on clinical need.
- **High-Tech Surgery:** Heart transplants (Christiaan Barnard 1967), hip replacements (John Charnley), keyhole laparoscopic surgery, and robotic systems.

ENQUIRY 04 pp. 9–10

Case Study: The Antibiotic Revolution & Penicillin

KEY QUESTION: *"How did an accidental laboratory mould transform into a wonder drug?"*

- **Fleming's Discovery (1928):** Alexander Fleming accidentally noticed penicillium notatum mould destroying staphylococcus bacteria at St Mary's Hospital, London.
- **Florey & Chain's Extraction (1938–1941):** Purified unstable penicillin at Oxford University; successfully tested on policeman Albert Alexander in 1941.
- **US Mass Production (WWII):** US pharmaceutical factories mass-produced penicillin for D-Day (1944), saving thousands of Allied soldiers and sparking modern antibiotics.

ENQUIRY 05 pp. 11–12

Case Study: Public Health & The Fight Against Lung Cancer

KEY QUESTION: *"How did the battle against lung cancer drive modern oncology?"*

- **Epidemiological Proof (1950):** Sir Richard Doll and Austin Bradford Hill proved the direct causal connection between cigarette smoking and lung cancer.
- **Government Public Health Laws:** Banned cigarette TV advertising (1965), introduced mandatory packet warnings, and banned smoking in public indoor venues (2007).
- **Advanced Oncology:** Radiotherapy (linear accelerators), chemotherapy, immunotherapy, and low-dose CT screening improving early detection and survival.

THEMATIC MATRIX pp. 2–11

Thematic Matrix: Longitudinal Factors & Synthesis

KEY QUESTION: *"How did universal healthcare and molecular biology redefine human life?"*

- **Welfare State & Government:** The post-war welfare consensus made national health a central state responsibility, ending Victorian medical inequality.
- **Science, Technology & Industry:** Pharmaceutical corporations, nuclear medicine, and international genome databases scaled treatments to the global population.
- **Modern Challenges:** Antibiotic resistance (superbugs like MRSA), aging populations with complex chronic conditions, and astronomical costs of novel therapies.

SPECIFICATION PAPER	SECTION B QUESTION ARCHITECTURE	ASSESSMENT WEIGHTING
Paper 1 Option 11 • Thematic Study	Q3: Similarity/Diff [4m] • Q4: Causation [12m] • Q5/Q6: Essay [	36 Raw Marks + 4 SPaG • 70% of Paper 1 T

ENQUIRY 01 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Ideas on Causes: Genetics, DNA & The Human Genome Project (c.1953–present)

KEY ENQUIRY QUESTION *How did unlocking the double helix and mapping the human genome revolutionize our understanding of disease causation?*

[1.1] Mendel, Heredity & The Chromosomal Mystery (c1900–1940s)

"Why did doctors assume proteins carried hereditary traits?"

In the early twentieth century, Germ Theory explained how external bacteria caused epidemic infections, but doctors could not explain why conditions like haemophilia ran in families. In 1900, scientists rediscovered **Gregor Mendel genetics** research from 1866. By cross-breeding thousands of pea plants, Mendel proved that physical characteristics were inherited in predictable mathematical ratios through invisible units later named genes.

Under high-powered laboratory compound microscopes, cellular **chromosomes observed** inside cell nuclei, but scientists could not decipher their complex chemical structure. For decades, researchers operated with an **unknown hereditary mechanism**, mistakenly assuming complex bodily proteins carried genetic instructions. Without understanding how genetic code was chemically stored, doctors remained entirely powerless to treat hereditary disorders or prevent inherited diseases.

[1.2] Rosalind Franklin & Photograph 51 (1951–1952)

"How did Photograph 51 prove DNA's helical structure?"

In 1951, physical chemist **Rosalind Franklin** joined King's College London to investigate the physical structure of DNA. A pioneer in **X-ray crystallography**, Franklin used high-powered X-ray beams to photograph shadow diffraction patterns of microscopic molecules. Working with PhD student Raymond Gosling, Franklin painstakingly exposed delicate crystallized DNA fibres to sixty-two continuous hours of radiation in a basement laboratory.

In May 1952, Franklin captured the historic **Photograph 51 (1952)**, revealing a distinct dark X-shaped pattern that proved DNA was a double-stranded helical spiral. In January 1953, Maurice Wilkins showed this image to James Watson without Franklin's knowledge or permission. Seeing the clear diffraction pattern provided Watson and Francis Crick with the crucial mathematical data needed to solve DNA's structure.

ENQUIRY 01 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

The DNA Revolution & The Human Genome Project

DECISIVE HISTORICAL MECHANISM Modern medicine shifted from treating external bacterial invaders to correcting internal cellular and genetic flaws.

[1.3] Watson & Crick's Double Helix Model (1953)

"How did base pairing explain hereditary replication?"

Using Franklin's crystallography measurements, Cambridge scientists James Watson and Francis Crick assembled the famous **Watson & Crick model** at the Cavendish Laboratory. Rather than performing wet chemical experiments, they constructed a physical wire-and-metal upright structure, proving that DNA formed a twisting **double helix structure** resembling a spiralling ladder held together by paired chemical bases.

Their decisive scientific discovery was **DNA base pairing**: adenine bonded strictly with thymine, while cytosine paired only with guanine. This explained how genetic information is copied during cell division, as unzipping strands act as templates to assemble identical replicas. Publishing in *Nature* in April 1953, they revealed the physical mechanism of heredity, revolutionizing biological science and modern medical pathology.

[1.4] The Human Genome Project (1990–2003)

"Why was mapping 3 billion base pairs a medical milestone?"

Deciphering DNA's molecular structure led directly to the launch of the **Human Genome Project** in 1990. Coordinated by international public bodies, including Britain's Wellcome Trust and Medical Research Council alongside American laboratories, this global scientific consortium set out to decode the complete instruction manual of human life. It was an extraordinary triumph of **international teamwork**, publishing all genomic findings freely online.

By April 2003, researchers had successfully **mapped 3bn base pairs** across twenty-five thousand human genes. Utilizing automated computer sequencing, scientists pinpointed the precise chromosomal locations of over four thousand faulty genes responsible for inherited conditions, including sickle cell anaemia, cystic fibrosis, and hereditary forms of breast and bowel cancer, transforming medical understanding of pathology.

[1.5] Genetic Screening & The Treatment Gap (2000s–Present)

"Why did understanding causes outpace finding cures?"

Decoding the human genome transformed clinical diagnostics, enabling routine **genetic screening** to identify individuals carrying hereditary disorders before clinical symptoms appear. For instance, **BRCA1/2 gene testing** allows women with family histories of breast cancer to assess their personal risk and elect for preventative mastectomies, transforming medicine from reactive emergency treatment into proactive, personalized genetic risk management.

However, a persistent **gene therapy lag** means that while identifying faulty chromosomes is now rapid and inexpensive, repairing or replacing defective genes inside living tissues remains extraordinarily difficult and costly. Doctors can diagnose thousands of inherited conditions with absolute precision, but effective clinical cures remain elusive, leaving a substantial gap between scientific diagnosis and practical bedside treatment.

ENQUIRY 02 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Lifestyle Factors & The Technological Revolution in Diagnosis (c.1900–present)

KEY ENQUIRY QUESTION *How did the shift toward lifestyle diseases and the rise of diagnostic scanning transform 20th-century medicine?*

[2.1] Wilhelm Röntgen & The Discovery of X-Rays (1895–1918)

"How did X-rays transform surgical trauma diagnosis in WWI?"

Before 1895, diagnosing internal fractures and gunshot wounds required agonizing exploratory surgery. This limitation ended when German physicist **Wilhelm Röntgen (1895)** discovered an invisible form of electromagnetic radiation while experimenting with cathode-ray tubes. Observing that these unknown rays penetrated flesh and cardboard but were absorbed by dense bone, Röntgen created the first diagnostic radiograph of his wife's hand.

During the First World War, Polish-French scientist Marie Curie equipped twenty **Marie Curie mobile vans** to bring radiographic equipment directly to frontline Casualty Clearing Stations. These mobile vehicles enabled military surgeons to achieve rapid **shrapnel detection**, locating bullets and jagged metal fragments lodged in soldiers' muscles before performing surgery, preventing catastrophic tissue damage and fatal battlefield infections.

[2.2] Blood Testing & Flexible Endoscopy (1930s–1960s)

"Why was fiber-optic endoscopy superior to exploratory surgery?"

In the mid-twentieth century, clinical diagnosis expanded from imaging bones to analyzing body chemistry. The introduction of routine **blood tests & typing** enabled doctors to identify anemia, monitor blood glucose levels in diabetics, and match compatible donors safely for blood transfusions. Laboratory **centrifuges & biopsy** analyses allowed pathologists to detect abnormal cellular changes and identify cancerous growths without major exploratory operations.

In the 1950s, optical physicists developed flexible fiber-optic glass cables capable of transmitting light and images around physical curves. Modern **fiber-optic endoscopes** allowed physicians to inspect digestive tracts, stomachs, and bronchial airways through natural bodily openings. This eliminated dangerous open-cavity exploratory surgery, allowing specialists to visualize internal bleeding, ulceration, and malignant tumours with minimal patient trauma.

ENQUIRY 02 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Non-Invasive Diagnostic Technology: From X-Rays to MRI

DECISIVE HISTORICAL MECHANISM Diagnostic imaging eliminated exploratory surgery and allowed early disease detection before symptoms manifest.

[2.3] Godfrey Hounsfield, CT Scans & MRI (1971–1980s)

"Why did CT scans transform internal soft-tissue diagnosis?"

While conventional X-rays clearly revealed dense skeletal structures, they could not depict soft internal organs, leaving brain tumours and internal bleeding invisible behind bone shadows. In 1971, British electrical engineer **Godfrey Hounsfield (1971)** resolved this problem by inventing Computerised Tomography, creating commercial **EMI CT body scanning** systems that revolutionized diagnostic imaging across modern hospital medicine.

CT scanners directed rotating X-ray beams through patients, using powerful computers to assemble thousands of cross-sectional measurements into detailed three-dimensional images of internal organs. By the 1980s, **MRI magnetic imaging** advanced diagnostics further, using strong magnetic fields and radio frequencies to image delicate brain tissue, spinal cords, and ligaments without exposing patients to ionizing radiation, achieving unprecedented diagnostic clarity.

[2.4] The Shift to Chronic & Lifestyle Diseases (c1950–1980s)

"What drove the shift from infectious to lifestyle diseases?"

In 1900, infectious epidemics like tuberculosis, pneumonia, and cholera caused most deaths in Britain. By the mid-twentieth century, clean municipal water infrastructure, compulsory childhood immunizations, and **antibiotics curb fevers**, bringing bacterial epidemics under historic control. Consequently, Britain experienced a **higher life expectancy**, with average lifespans rising past seventy-five years for both men and women across the nation.

However, living decades longer generated a profound epidemiological shift: modern mortality became dominated by **cancers & heart disease**. Medical researchers proved that chronic illnesses were driven by daily lifestyle choices, including cigarette smoking, high-sugar diets, physical inactivity, and excessive alcohol consumption, compelling the medical profession to shift attention from treating acute infections to managing long-term lifestyle conditions.

[2.5] State Lifestyle Campaigns & Prevention vs Cure (1980s–Present)

"Why do governments favour lifestyle prevention over treatment?"

Treating advanced chronic conditions placed enormous financial pressure on the NHS, prompting the British government to embrace preventive public health campaigns. Parliament introduced statutory safety measures, including mandatory **1983 Seatbelt legislation** to slash motor vehicle trauma, alongside nationwide public health initiatives like the **sugar tax & Change4Life** campaign, educating families to improve daily nutrition and reduce childhood obesity.

The most impactful statutory intervention was the **2007 Public smoking ban**, prohibiting smoking in all enclosed workplaces and public venues across England. By eliminating secondhand smoke in pubs and restaurants, the law reduced hospital admissions for heart attacks and established the modern principle that government legislation and preventive education save vastly more lives than expensive hospital technology.

ENQUIRY 03 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

The Search for Magic Bullets, High-Tech Treatments & The Birth of the NHS (1909–1948)

KEY ENQUIRY QUESTION *How did synthetic magic bullets and the foundation of the NHS dismantle medical inequality in Britain?*

[3.1] Paul Ehrlich & Salvarsan 606 (1906–1909)

"Why was Salvarsan 606 termed the first "magic bullet"?"

By 1900, bacteriology had identified specific disease-causing microbes, but doctors lacked chemical treatments that could kill bacteria inside living patients without causing fatal poisoning. In Germany, scientist **Paul Ehrlich research** pioneered chemotherapy, introducing the revolutionary **magic bullet concept**—a synthetic chemical compound that would seek out and destroy specific microbes while leaving surrounding bodily tissues completely unharmed.

Working with Japanese bacteriologist Sahachiro Hata, Ehrlich systematically tested hundreds of arsenic chemical derivatives. In 1909, their 606th experimental compound successfully destroyed the spirochaete bacterium responsible for syphilis. Marketed internationally as **Salvarsan 606 (1909)**, this became the world's first synthetic chemical cure, proving scientifically that targeted chemical compounds could eradicate specific internal bacterial infections inside the living human body.

[3.2] Gerhard Domagk & Prontosil (1932–1935)

"How did Prontosil prove synthetic drugs could kill microbes?"

While Salvarsan 606 cured syphilis, it was ineffective against common bacterial killers like blood poisoning, scarlet fever, and pneumonia. In 1932, German pathologist **Gerhard Domagk (1932)** discovered that a synthetic red leather dye named **Prontosil red dye** killed lethal streptococcal bacteria in laboratory mice without damaging their healthy internal organs or causing fatal toxicity.

In 1935, Domagk treated his own daughter with Prontosil when she faced fatal blood poisoning from an infected embroidery needle, achieving a rapid clinical recovery. Researchers at the Pasteur Institute discovered its active ingredient, sulphonamide, triggering the global **sulfa drug revolution**. Inexpensive, mass-produced sulphonamides slashed maternal puerperal fever mortality and saved millions of civilian lives from deadly bacterial infections.

ENQUIRY 03 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Magic Bullets vs Broad-Spectrum Antibiotics

DECISIVE HISTORICAL MECHANISM Key Distinction: Magic bullets are synthetic chemical compounds; antibiotics are natural antimicrobial substances produced by microorganisms.

[3.3] The Beveridge Report & Wartime Reconstruction (1942)

"How did WWII unity create momentum for state healthcare?"

Before 1948, healthcare in Britain was deeply unequal. Under the 1911 National Insurance Act, only insured male workers received subsidized GP care, leaving wives, children, and the elderly dependent on charity hospitals or personal savings. The shared dangers of the Blitz, mass evacuation, and food rationing created widespread public solidarity and demand for egalitarian state welfare reform across all classes.

In December 1942, economist William Beveridge published the landmark **1942 Beveridge Report**, proposing a comprehensive system of social insurance to slay the **Five Giant Evils**: Want, Disease, Ignorance, Squalor, and Idleness. The report became an instant bestseller, laying the architectural foundation for a **post-war welfare state** that guaranteed medical care as a fundamental universal right for all citizens.

[3.4] Aneurin Bevan & The Launch of the NHS (1945–1948)

"How did Bevan compromise to persuade resistant doctors?"

In 1945, the newly elected Labour government appointed former Welsh miner **Aneurin Bevan (1948)** as Minister of Health. Bevan was determined to establish a state-run National Health Service founded on **free universal healthcare**, funded entirely through central taxation and available to every resident without charge at the point of delivery, transforming medical access across post-war Britain.

Over ninety percent of doctors opposed state control through the British Medical Association, fearing the loss of clinical independence. Bevan secured a crucial **BMA doctor compromise** by allowing hospital consultants to retain lucrative private patients, famously remarking he had "stuffed their mouths with gold." On 5 July 1948, the NHS launched, providing comprehensive medical care to millions.

[3.5] High-Tech Care, Surgical Milestones & Escalating Costs (1950s–Present)

"Why has modern medical success created financial strain?"

Following the creation of the NHS, British hospitals embraced high-tech surgery. In 1962, Sir John Charnley perfected low-friction **hip replacement surgery**, restoring mobility to hundreds of thousands of elderly patients. Simultaneously, mechanical **kidney dialysis units** cleansed toxic blood in patients suffering renal failure, while organ transplants and minimally invasive keyhole operations drastically accelerated patient recovery and reduced surgical trauma.

However, clinical success created severe financial challenges. An aging population, public expectations of modern pharmaceuticals, and expensive diagnostic technologies generated **rising NHS cost pressures**. Managing escalating hospital budgets for cutting-edge treatments while maintaining Bevan's founding principle of free universal access at the point of delivery remains the primary structural dilemma of contemporary British healthcare.

ENQUIRY 04 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Case Study 1: The Antibiotic Revolution: Fleming, Florey & Chain and Penicillin (1928–1945)

KEY ENQUIRY QUESTION *How did an accidental laboratory mold transform into humanity's first mass-produced antibiotic wonder drug?*

[4.1] Alexander Fleming's Accidental Discovery (1928–1929)

"Why could Fleming not turn mould into a clinical medicine?"

During the First World War, Scottish bacteriologist **Alexander Fleming (1928)** witnessed thousands of soldiers die from infected wounds, observing that carbolic antiseptics damaged white blood cells without reaching deep bacterial infections. In September 1928, returning from holiday to his untidy laboratory at St Mary's Hospital, London, Fleming examined discarded culture plates of pathogenic bacteria piled on his workbench.

Fleming noticed a drifting spore of **Penicillium notatum mould** had contaminated a petri dish, surrounded by a distinct **staphylococcus halo** where bacteria had dissolved completely. Fleming named the non-toxic fluid "penicillin." Although publishing his findings in 1929, Fleming could not extract or chemically stabilize pure penicillin, leaving it an unexploited laboratory curiosity for nearly a decade.

[4.2] Florey, Chain & The Oxford Team (1938–1940)

"How did Norman Heatley solve the purification challenge?"

In 1938, Australian pathologist Howard Florey and German-Jewish refugee biochemist Ernst Chain re-examined Fleming's paper at Oxford University. Leading an interdisciplinary research team of chemists and bacteriologists, **Florey & Chain (Oxford)** recognized penicillin's revolutionary potential, securing essential research funding from the British Medical Research Council and the American Rockefeller Foundation to investigate its clinical properties.

The team faced the enormous technical obstacle of **purifying active penicillin** in usable therapeutic quantities. Oxford technician Norman Heatley designed ingenious laboratory equipment, assembling a makeshift **Norman Heatley apparatus** of milk churns, hospital bedpans, and glass flasks to brew culture fluid in university basements, freeze-drying the delicate chemical into a stable brown therapeutic powder that could be safely injected into living patients.

ENQUIRY 04 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

The Industrial Scale of Penicillin: Florey, Chain & WWII

DECISIVE HISTORICAL MECHANISM Industrial mass-production, government wartime finance, and US chemical technology turned penicillin into a reality.

[4.3] Mice Trials & The Tragic Case of Albert Alexander (1940–1941)

"Why was Alexander's trial both a success and a tragedy?"

In May 1940, the Oxford researchers conducted their decisive **1940 Lethal mice trial**. Eight mice were injected with fatal doses of streptococcus bacteria; four were treated with penicillin while four remained untreated. Within twenty-four hours, the untreated mice died, whereas the four penicillin-treated mice recovered completely, providing indisputable experimental evidence that penicillin cured fatal bacterial infections in living animals.

In February 1941, the team treated Oxford policeman **Albert Alexander (1941)**, who was dying from severe facial blood poisoning. Alexander improved dramatically after receiving penicillin, but supplies were so scarce that doctors recycled the drug from his urine. When stocks ran out, Alexander suffered a tragic **supply exhaustion death**, demonstrating that clinical survival required industrial mass-production.

[4.4] Peoria & US Industrial Mass Production (1941–1944)

"How did US industrial fermentation enable mass production?"

With British chemical plants committed to wartime explosives and vulnerable to Luftwaffe bombing, Florey and Heatley travelled to the United States in July 1941. Working with the US Department of Agriculture in Peoria, Illinois, researchers discovered that corn-steep liquor and a cantaloupe mould strain yielded two hundred times more penicillin, securing substantial **US government funding** for industrial manufacturing across major pharmaceutical corporations.

American pharmaceutical corporations constructed massive industrial **deep fermentation tanks**, pumping sterile air through thousands of gallons of nutrient broth to mass-produce pure antibiotic medicine. By June 1944, American factories supplied **2.3m D-Day doses** to Allied armed forces, preventing gangrene, eliminating surgical amputations, and ensuring thousands of wounded soldiers survived battlefield trauma.

[4.5] D-Day Allied Victory & The Superbug Threat (1944–Present)

"Why did Fleming warn that antibiotic overuse causes resistance?"

Penicillin transformed military surgery, securing **Allied troop survival** across the European theatre and slashing wound infection fatalities during the Normandy campaign. Following their joint **1945 Nobel Prize**, Fleming, Florey, and Chain were celebrated globally. Available freely through the NHS, penicillin wiped out deadly streptococcal throat infections, pneumonia, and syphilis, launching the twentieth-century golden age of antibiotic chemotherapy across Britain.

However, widespread clinical over-prescription and agricultural misuse created a dangerous modern crisis: antimicrobial resistance. Bacterial strains evolved genetic defenses, creating the lethal **MRSA superbug threat** in modern hospital wards. This demonstrated how revolutionary biomedical breakthroughs require continuous scientific innovation, strict prescription stewardship, and hospital infection control to preserve their life-saving clinical efficacy for future generations.

ENQUIRY 05 • PART I

PEARSON EDEXCEL GCSE (9–1) HISTORY • PAPER 1 (1HI0/11)

Case Study 2: Public Health & The Fight Against Lung Cancer (c.1950–present)

KEY ENQUIRY QUESTION *How did the battle against lung cancer drive modern high-tech oncology and permanently dismantle government laissez-faire?*

[5.1] The Diagnostic Challenge: Why Early Lung Cancer Eludes Detection

> *"Why did early doctors fail to link smoking to lung cancer?"*

In the twentieth century, lung cancer emerged as the deadliest malignant epidemic in Great Britain, driven by the mass popularity of cigarette smoking. Early diagnosis remained extraordinarily difficult because human lungs possess **few pain receptors** inside internal alveolar tissues. Patients do not feel developing tumours expanding, mistaking early symptoms for a harmless smoker's cough or mild breathlessness.

Consequently, malignant tumours develop through silent **asymptomatic growth** for months or years without causing clinical distress. By the time patients experience acute warning signs like persistent chest pain or coughing blood, the disease has reached **late Stage 3/4 detection**. At this advanced stage, cancer cells have metastasized to the lymph nodes and other organs, rendering curative surgery impossible.

[5.2] High-Tech Diagnostic Scans: CT, PET-CT & Bronchoscopy

> *"How did statistical epidemiology prove the causal link?"*

Conventional 2D chest X-rays were inadequate for detecting early lung cancer, as small tumours remained hidden behind the heart and ribcage shadows. Modern diagnosis was revolutionized by **high-resolution CT scans**, which rotate around the patient's thorax to assemble hundreds of digital cross-sections, pinpointing microscopic nodules long before they cause noticeable physical symptoms or severe respiratory distress.

Oncology centers combine CT with advanced **PET-CT tracer imaging**, injecting radioactive glucose that lights up fast-growing cancer cells to show whether tumours are active and spreading. Specialists then perform a **bronchoscopy & biopsy**, sliding a thin flexible camera down the airways to extract cellular tissue, confirming the precise histological subtype of cancer to design targeted clinical treatments.

ENQUIRY 05 • PART II

TURNING POINTS & HISTORICAL CONSEQUENCES

Public Health & Lifestyle: The Anti-Smoking Crusade

DECISIVE HISTORICAL MECHANISM Government action transitioned from sanitary infrastructure (sewers) to public education and behavioral regulation.

[5.3] Precision Interventions: Keyhole Surgery & CyberKnife Radiotherapy

"Why did government action encounter strong tobacco lobbying?"

Historically, surgical removal of lung tumours required major open thoracotomy, a high-risk operation cutting through chest muscles and spreading ribs that caused severe pain and prolonged recovery. Modern thoracic surgeons instead utilize **VATS keyhole lobectomy**, inserting microscopic cameras and surgical instruments through tiny incisions, meaning traumatic **chest cracking avoided** and patient recovery times drastically reduced in hospital wards.

For elderly patients or those too frail for surgery, modern oncology utilizes advanced radiotherapy systems like the **CyberKnife pinpoint rays**. This robotic machine fires targeted beams of high-dose radiation from hundreds of angles directly into the tumour while tracking patient breathing, destroying malignant cancer cells while preserving surrounding healthy lung tissue from destructive radiation damage.

[5.4] Targeted Genetic Oncology: Gene Sequencing & Immunotherapy

"Why was passive smoking crucial in banning public smoking?"

In the twenty-first century, decoding the human genome transformed modern lung cancer treatment from blunt chemotherapy into precision genetic oncology. Pathologists perform routine **tumour DNA sequencing** on biopsy tissue to identify specific genetic mutations that stimulate tumour growth, such as EGFR mutations, allowing oncologists to personalize therapy to each individual patient's unique cellular profile and cancer characteristics.

Doctors prescribe advanced **targeted pharmacotherapy** drugs that block the specific molecular proteins feeding the tumour, halting cancer progression without destroying healthy blood cells. Alongside this, **targeted immunotherapy** medications unmask cancer cells, training the patient's own immune system to identify and destroy malignant tumours, significantly extending life expectancy for patients with advanced lung cancer across NHS hospitals.

[5.5] Government Statutory Action: Legislation & Prevention Campaigns

"Why does low cancer survival prove prevention beats cure?"

While high-tech oncology improved survival rates, the British government recognized that prevention was vastly more effective than attempting to cure advanced disease. The decisive legislative breakthrough was the landmark **2007 Public smoking ban**, prohibiting smoking in all enclosed workplaces, restaurants, and public transport across England, virtually eliminating deadly secondhand smoke exposure and encouraging thousands of citizens to quit permanently.

Parliament reinforced prevention through the **2016 Plain packaging** law, replacing branded logos with standardized drab olive boxes covered in graphic health warnings. Looking ahead, statutory initiatives like the proposed **smoke-free generation law** aim to raise the legal smoking age annually, proving that government legislation and public health education remain the most decisive weapons against lung cancer.

Master Knowledge Retrieval Quiz

Modern Medicine & High-Technology Healthcare • c1900–present (30 Essential Recall Questions)

Standard Recall Protocol: Partner Quiz or Solo Practice. Test your rapid chronological, factual, and conceptual recall across all 5 enquiries. All questions are cross-referenced directly to paragraph locations in this textbook; zero answers are printed to maintain active retrieval integrity.

Enquiry 01: Ideas on Causes: Genetics, DN...

pp. 2–3 • Ref [1.1]–[1.5]

1. How did Dr John Snow prove that the 1854 Soho cholera outbreak was water-borne?
2. Why did the workers at the Broad Street Lion Brewery survive the 1854 cholera outbreak?
3. What crisis in July 1858 forced Parliament to authorize London's underground sewer system?
4. Who was the civil engineer who designed and built London's intercepting sewer network?
5. Why was the 1875 Public Health Act far more successful than the 1848 Public Health Act?

Enquiry 02: Lifestyle Factors & The Techno...

pp. 4–5 • Ref [2.1]–[2.5]

6. What does DNA stand for?
7. In what year did Watson and Crick publish their double-helix model of DNA?
8. Which scientist took the crucial X-ray diffraction image known as Photograph 51?
9. Why could Pasteur's Germ Theory not explain illnesses like haemophilia and cystic fibrosis?
10. Which 19th-century monk's plant-breeding experiments established the foundational laws of heredity?

Enquiry 03: The Search for Magic Bullets, ...

pp. 6–7 • Ref [3.1]–[3.5]

11. In what year did Wilhelm Röntgen discover X-rays?
12. Why was the discovery of X-rays in 1895 immediately significant for surgery?
13. Who pioneered mobile radiology ambulances ("petites Curies") during WWI?
14. Which British engineer invented the CT scanner at EMI in 1971?
15. How did the CT scanner (invented in 1971) improve upon conventional 2D X-rays?

Enquiry 04: Case Study 1: The Antibiotic R...

pp. 8–9 • Ref [4.1]–[4.5]

16. What term did Paul Ehrlich coin for chemicals that destroy bacteria without harming human tissue?
17. What disease was cured by Salvarsan 606 in 1909?
18. Which Japanese scientist assisted Paul Ehrlich in testing arsenic compounds to discover Salvarsan 606?
19. Who discovered that the red dye Prontosil cured bacterial infections in 1932?
20. What active antibacterial chemical group was found inside Prontosil?

Enquiry 05: Case Study 2: Public Health & ...

pp. 10–11 • Ref [5.1]–[5.5]

21. In what year did Alexander Fleming discover penicillin at St Mary's Hospital?
22. What type of bacteria was being cultured in the petri dish when penicillin was discovered?
23. What was the scientific name of the mold that contaminated Fleming's dish?
24. Who led the Oxford University research team that purified penicillin starting in 1938?
25. What ingenious technician built improvised laboratory apparatus using bedpans and dairy churns at Oxford?

Synoptic Synthesis & Historical Causation

pp. 2–11 • Whole-Era Recall

26. **[Synoptic]** Explain the single biggest obstacle to medical progress across this era.
27. **[Synoptic]** Which institution or technological breakthrough triggered the greatest shift in clinical practice?
28. **[Synoptic]** How did contemporary beliefs about the causes of disease influence the treatments administered?
29. **[Synoptic]** Assess the extent of continuity between the start and end of this chronological period.
30. **[Synoptic]** Identify one individual whose contributions laid the groundwork for subsequent breakthroughs.